

Fact-check of the 0G to 6G Generation Table

Mini Project 1.3: annotated comparison against additional sources

Sources [S1] to [S8] are listed in the full report (report_1_3.pdf). Compare the "Table claim" column with your Section 1.2 table.

Generation	Table claim	What sources say	Discrepancy and reason
2G (GSM, GPRS, EDGE)	about 9.6 kbps (GSM); tens to hundreds of kbps with GPRS/EDGE	GSM 9.6 kbps typical; GPRS 171.2 kbps theoretical but about 58 kbps in practice [S1, S2]; EDGE up to 384 kbps, 120–384 kbps in the field [S1]	Theoretical peaks assume all time slots and the best coding scheme for one user. One source lists GPRS up to 56 kbps and EDGE up to 256 kbps [S3], reflecting handset slot limits.
3G (UMTS, HSPA)	about 384 kbps to 2 Mbps	UMTS 384 kbps to 2 Mbps, shared by all active users in a 5 MHz cell, falling to 384 kbps with two users [S2]. HSPA theoretical 42 Mbps but averages 1–3 Mbps [S1]; another source lists 7.2 Mbps and 42 Mbps with dual carrier [S3]	Peak rates depend on sharing and release, so "3G" spans a factor of 100 or more.
4G (LTE)	about 100 Mbps (LTE)	100 Mbps theoretical, 3–10 Mbps average and 20+ Mbps peak in real use [S1]; early LTE listed at up to 75 Mbps [S3]	Theory versus field measurement. S1 dates from 2014, so verify against a current source.
5G (NR)	about 10–20 Gbps peak	ITU minimum: 20 Gbps DL and 10 Gbps UL peak, 100 Mbps user-experienced DL, 1 ms URLLC latency, 10^6 devices per km ² [S4, S5]. M.2083: peak expected to reach 10 Gbps, up to 20 Gbps in certain conditions [S6]	These are ITU evaluation requirements, not deployment performance. The user-experienced rate is 200 times smaller than the peak. Add a field-measurement source.
6G	about 1 Tbps (vision)	IMT-2030 targets reported as 50, 100 and 200 Gbps by scenario; 1 Tbps is the most common research target; user-experienced 300–500 Mbps; latency 0.1–1 ms [S7]	6G is not standardized or deployed; label these figures as targets.